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Adjustments of posture and viewing parameters of the eye to changes in the screen height of the visual display terminal

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Keywords: Visual display terminal (VDT); Screen height; Posture; Eye position; Ocular surface area (OSA).

To understand the motions caused by the interaction among the different body parts, adjustments of both eye position and body posture to screen height settings of 80, 90, 100, 110 and 120 cm were determined in 10 subjects. The subjects engaged in a non-keyboard, interactive computer game requiring constant visual monitoring. Changes in body positions were determined by video image analysis for the posture parameters and by video frame analysis for the eye parameters. Analysis of variance and correlation analysis showed that neck angle, thoracic bending and vertical eye position were significantly affected by changes in the screen height of the VDT. The study of the interrelationship of eye position and body posture suggested that changes in body positions served to complement the eye position in attaining a better view of the visual target. Viewing angle was observed to be decided mainly by inclination of the neck and the eye. Thoracic bending was also noted to contribute in setting the viewing angle, although to a lesser extent. On the other hand, viewing distance and trunk inclination showed significant correlation.

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1. Introduction

The introduction of visual display terminals (VDT) in the workplace has led to numerous reports of health disorders from its use. The disorders are predominantly referable to the eye and the musculoskeletal system (Hünting *et al.* 1981, Ardnt 1983, Knave *et al.* 1985, Gobba *et al.* 1988, Bergqvist *et al.* 1995). Factors believed to contribute to the development of these disorders include static position assumed during work, increased work intensity and improper workstation design. As a consequence, the spawning of many guidelines aimed at preventing adverse effects in workers using visual display terminals were observed.

Recommendations for visual comfort favour a downward eye inclination thereby advocating workstation setting below the level of the eye (Kroemer and Hill 1986, Heuer *et al.* 1991, Taptagaporn and Saito 1993). Workstation designs were proposed to offset musculoskeletal problems (Grandjean 1984, Cakir *et al.* 1980). These guidelines, however, apply to conventional VDT workstations where the CRT screen is used in an office setting or in the performance of a desktop task.

The increasing popularity of various screen sizes and height locations compounds the dilemma of posture and eye position during work. The recent advent of large-size

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monitors for convenient use of graphic user interfaces (GUI), such as icons and symbols, may induce an upward gaze and a more erect body posture. Similar body positions of workers may be noted in traffic control centres and nuclear power facilities because of the upward location of the screen. The use of flat panel displays (FPD), on the other hand, may bring about a downward gaze and a bent posture.

The integration of the VDT in many workplaces creates a new dimension to the increasing number of ergonomic concerns with VDT use. Constrained postures are imposed among VDT users performing tasks that require sustained visual monitoring (Hünting *et al.* 1980) such as in vigilance tasks observed in air traffic controls, medical and industrial monitoring, and military target spotting (Craig and Davies 1991). The introduction of various non-keyboard input devices, e.g. mice, pucks and joysticks, in workstations for graphic and engineering designs and programming may further restrict free movement among VDT users. Further, VDT tasks, while using these non-keyboard input devices such as pointing, dragging and selection, require more concentrated fixation of the operator's eyes on the screen that would often result in constrained postures.

To better understand the outcomes of VDT use, it is vital to comprehend how the different body parts adapt to a complex problem. It is the aim of this study to determine the ergonomic problems in VDT work with non-keyboard input devices. Postural reaction to changes in the height of the screen while performing a non-keyboard task was determined. Body and eye positions affected by screen height modifications were ascertained. Mutually related body and eye movements were also identified.

2. Subjects and method

2.1. *Subjects*

Ten subjects (8 male and 2 female) participated in the experiment. Mean age was 36·1 ± 8·8 years. Mean height was 168·8 ± 5·5 cm.

2.2. *Experimental workstation*

The selection of the five screen height settings, i.e., 80, 90, 100, 110 and 120 cm, used in the experiment was based on the recommended table height of 67 cm (Japanese Industrial Standard S1010-1978 1993). The most prevalent CRT size of 14 in. was selected for the experiment. The central processing unit (CPU) and the CRT monitor placed on top of a table of height 67 cm resulted in a screen height of 100 cm (from the floor to the middle of the screen). Change of screen height was made at 10-cm increments.

To better evaluate the interaction between the screen and the operator, the mouse table height was fixed. The table height upon which the mouse rests was constant for all subjects at 67 cm. The mouse table is detached from the adjustable screen table.

Prior to the start of the experiment, the subjects were asked to adjust the chair until the forearm was parallel to the ground when handling the mouse. This forearm position prevents extreme hand and arm posture responsible for local musculoskeletal discomfort (Chaffin and Andersson 1991, Hagberg *et al.* 1995). The back support was adjusted to desired levels. Thereafter, the chair setting remained unchanged throughout the experiment. These experimental conditions are considered to be similar to the actual VDT workstations where larger or smaller sizes of screens were used on traditional desks.

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The illumination levels were 230 lx on the mouse table and 120 lx on the CRT. The luminance level of the brightest area of the screen was 79 cd/m^2 while that of the darkest area was 11 cd/m^2.

2.3. *Task*

The subjects were asked to engage in an interactive computer game. The game required constant visual monitoring—similar to a vigilance task—to detect signals that would determine the operator's next action. To purely extract the effects of the screen height on posture and eye position and to highlight human-screen interaction, only the mouse was used as the input device in this experiment. Moreover, keyboard operation sometimes requires human-keyboard interaction that may confound the analysis.

The subjects performed the VDT task for 3 min at each screen height to allow adaptation to the workstation. Posture and eye position were evaluated at the end of the allotted time.

Results of long duration posture evaluation of 4 out of the 10 subjects provided a basis for the observation period in this experiment. Under the same experimental conditions as in the present study, the posture of the subjects was observed while performing the task for 50 min (Japan Industrial Safety and Health Association 1985) at the different screen height settings. The body positions measured were the same as those described in § 2.4. The mean coefficient of variation of the different body positions evaluated during the long duration experiment was 3·1% (SD = 2·4%; range = 0·7–9%).

Rest was introduced in between height changes. The subjects were also asked to stand away from the experimental workstation in between height changes.

2.4. *Posture analysis*

2.4.1. *Body parameters*: Five markers were put on the subjects, the location of which

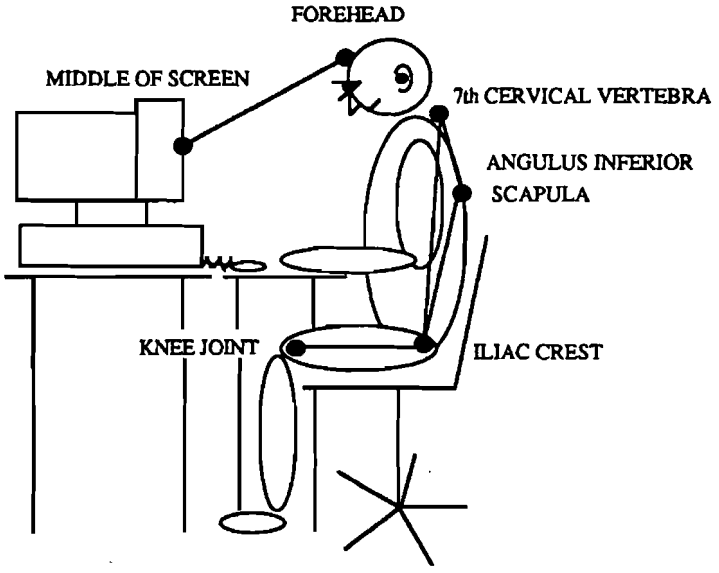


Figure 1. Location of markers and measurement points for body posture.

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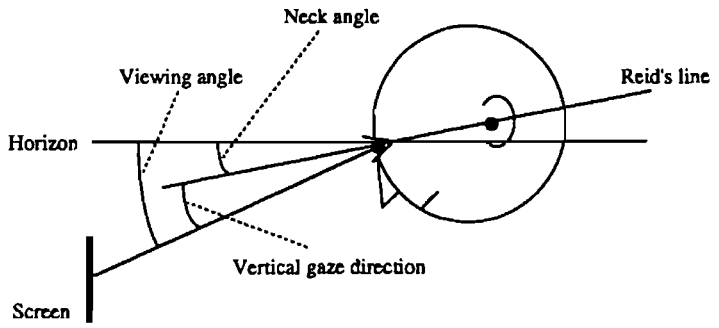


Figure 2. Angles measured in determining eye position.

were the following: (1) forehead; (2) seventh cervical vertebra (C7); (3) inferior angle of the scapula; (4) iliac crest; and (5) knee joint. One marker was placed at the middle of the screen (figure 1). Body positions at different screen height settings were captured by a video camera from the left profile of the subject. Posture was analysed using a video image analyser and by frame analysis of the videotape recording.

The following angles were measured:

1. *Neck angle*: based on the Reid's line referenced to the horizontal plane. The Reid's line is defined by a line connecting the outer canthus of the eye and the centre of the outer canal of the ipsilateral ear (figure 2). This study used the Reid's line because it is easier to measure from the video image than the Frankfurt plane—defined by the horizontal plane at the level of the upper edge of the opening of the external auditory meatus and the lower border of the orbital margin (ISO 1995). The Frankfurt plane is defined anatomically, thus making its determination by video image difficult. Earlier findings showed that the Reid's line is approximately 10° above the Frankfurt plane.
2. *Thoracic bending*: angle formed by C7, angulus inferior scapula and iliac crest. A smaller angle corresponds to a relatively more kyphotic spine.
3. *Trunk inclination*: angle formed by C7 and iliac crest referenced against the horizontal plane. Reclining positions would assume a value of more than 90°.
4. *Pelvic angle*: angle formed by C7, iliac crest and knee joint. The angle corresponds to the degree of hip flexion.

2.4.2. *Viewing parameters of the eye*: Vertical gaze direction and ocular surface area (Sotoyama *et al.* 1995) were estimated by measuring the palpebral fissure width from the video image of the eye. Correlation between palpebral fissure width and both vertical gaze direction and ocular surface area have been established to be strong with correlation coefficients greater than 0·95. The viewing angle is defined as the angle formed by the horizontal plane and the line between the eye and the visual target. Contributions of both neck inclination and eye inclination were taken into account to determine the viewing angle (figure 2). Viewing distance was determined by measuring the distance between the marker at the level of the middle of the screen and that on the forehead.

2.5. *Statistical analysis*

Correlation analysis, one-way ANOVA and paired *t*-test were used to evaluate the results. Statistical tests with $p < 0.05$ were considered to be significant.

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3. Results

3.1. *Body posture and screen height*

Results showed that neck angle and thoracic bending were significantly affected by changes in screen height (table 1). Mean levels of neck angle at different screen heights demonstrated that the neck became more erect as the screen position became higher. The trend in the bending of the trunk similarly showed that the back straightened as the screen height was raised.

Trunk inclination and the pelvic angle did not significantly change with the change in the screen height. None the less, it is interesting to note that the mean trunk inclination at all screen height settings showed that the subjects preferred a backward-leaning position. A schematic representation was made to understand better the interaction of the body positions and viewing angle at different height settings (figure 3). The results of the changes in viewing angle will be discussed in the next section.

Comparison of the correlation coefficients suggested a stronger correlation between neck angle and screen height ($r = 0{\cdot}740$; slope $= 0{\cdot}566$) than between the latter and thoracic bending ($r = 0{\cdot}568$; slope $= 0{\cdot}162$) (figures 4 and 5). The neck moved more and was seemingly more vulnerable as the screen height was changed.

3.2. *Viewing parameters of the eye and screen height*

Changes in the screen height induced significant changes in the vertical gaze direction, ocular surface area and viewing angle (table 2). The eye followed the change in the

Table 1. Mean levels of body positions at different screen heights.

Screen height ($n = 10$)	Parameters (Mean ± SD)			
	Neck angle (°)	Thoracic bending (°)	Trunk inclination (°)	Pelvic angle (°)
80 cm	4·7 ± 9·1	116·8 ± 4·2	99·4 ± 10·3	115·9 ± 11·1
90 cm	11·5 ± 8·2	117·4 ± 3·4	99·9 ± 12·6	115·6 ± 13·3
100 cm	15·5 ± 7·9	121·4 ± 1·6	95·5 ± 7·1	113·0 ± 7·5
110 cm	22·9 ± 6·3	120·2 ± 2·2	98·7 ± 9·9	115·6 ± 8·5
120 cm	27·3 ± 6·4	123·5 ± 4·3	99·4 ± 9·4	105·9 ± 34·3
For 50 observations	16·4 ± 10·9	119·9 ± 4·1	98·6 ± 9·7	113·2 ± 17·6
ANOVA	$p < ·001$	$p < ·001$	$p = ·89$	$p = ·98$†

†Kruskal-Wallis test.

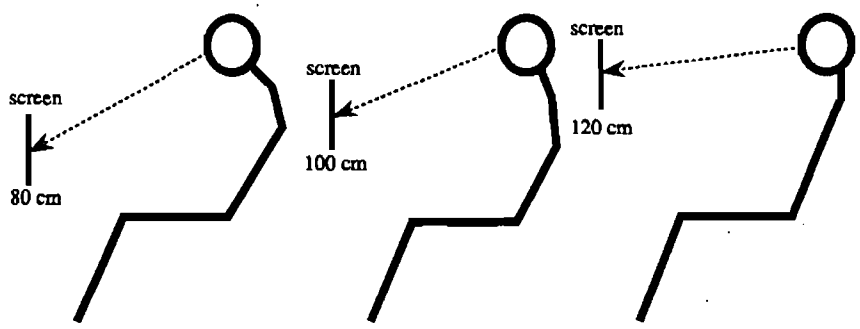


Figure 3. Schematic representation of body posture at different screen heights.

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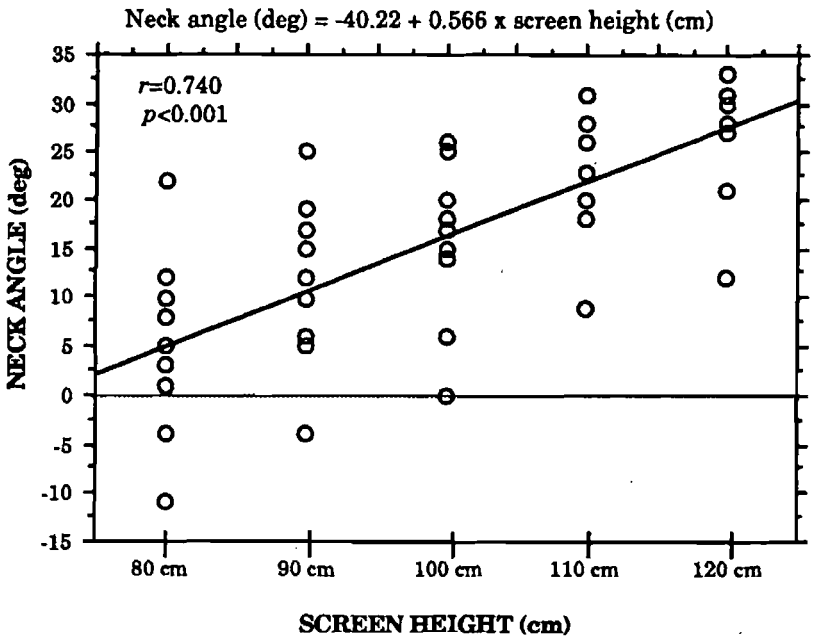


Figure 4. Relationship between screen height and neck angle.

Table 2. Mean levels of eye parameters at different screen heights.

Screen height ($n = 10$)	Parameters (Mean ± SD)			
	Viewing distance (cm)	Ocular surface area (cm^2)	Vertical gaze direction (°)	Viewing angle (°)
80 cm	60·6 ± 7·5	1·6 ± 0·2	−43·2 ± 9·9	−38·5 ± 12·2
90 cm	56·0 ± 6·7	1·8 ± 0·1	−39·0 ± 8·2	−27·5 ± 10·9
100 cm	54·6 ± 8·0	2·1 ± 0·1	−32·7 ± 7·2	−17·2 ± 11·5
110 cm	54·7 ± 7·2	2·1 ± 0·1	−33·0 ± 10·4	−10·1 ± 10·2
120 cm	55·0 ± 7·6	2·3 ± 0·2	−27·9 ± 9·4	−0·6 ± 12·1
For 50 observations	56·2 ± 7·5	2·0 ± 0·4	−35·1 ± 10·3	−18·8 ± 17·3
ANOVA	$p = ·34$	$p < ·01$	$p < ·01$	$p < ·001$

height of the screen. The elevation of both gaze direction and viewing angle and the increase in the ocular surface area were noted as the height of screen was raised.

The viewing distance was maintained as the height of the screen was altered. The longest distance was noted, though, at the screen height of 80 cm.

3.3. *Relationship of body positions and viewing parameters of the eye*

Correlation analysis showed that the relationship among the different body and eye positions may be grouped according to the plane of movement as shown in table 3. All parameters moving along the vertical plane in the table, the neck angle and thoracic bending correlated significantly with the vertical gaze direction and viewing angle. Similarly, parameters moving along the horizontal plane, i.e. viewing distance, trunk inclination and pelvic angle, demonstrated significant correlation.

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Table 3. Interrelationship of body and eye parameters.

Correlation matrix	Neck angle (°)	Thoracic bending (°)	Trunk inclination (°)	Pelvic angle (°)	Viewing distance (°)	Ocular surface area (cm^2)	Vertical gaze direction (°)	Viewing angle (°)
Neck angle (°)	1·000	–	–	–	–	–	–	–
Thoracic bending (°)	0·358	1·000	–	–	–	–	–	–
Trunk inclination (°)	−0·123	−0·006	1·000	–	–	–	–	–
Pelvic angle (°)	−0·184	−0·216	0·534***	1·000	–	–	–	–
Viewing distance (cm)	−0·212	−0·146	0·500***	0·500***	1·000	–	–	–
Ocular surface area (cm^2)	0·327	0·657***	0·256	0·036	−0·269	1·000	–	–
Vertical gaze direction (°)	0·327	0·657***	0·254	0·036	−0·270	1·000	1·000	–
Viewing angle (°)	0·828***	0·655***	0·073	−0·096	−0·295	0·801***	0·801***	1·000

2-tailed significance: $^{**}p < ·01$; $^{***}p < ·001$.

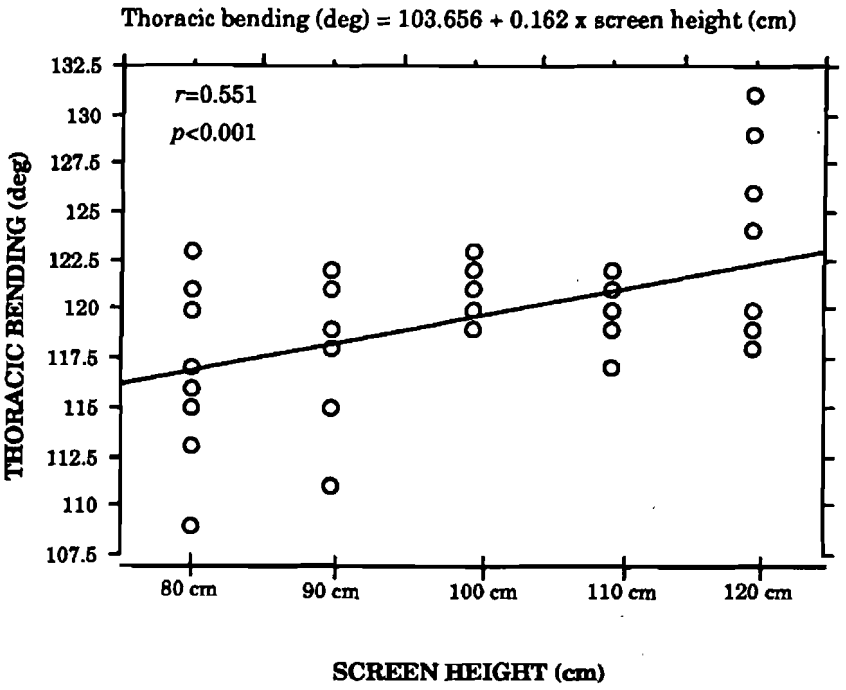


Figure 5. Relationship between screen height and thoracic bending.

The relationship between neck angle and viewing angle is invariably linear. As the viewing angle became lower, neck flexion was increased. This is not surprising considering that the eyes are housed in the head and are in close proximity to all the pivot points for head and neck movement. The bending of the thorax also changed, although not of the same magnitude as the neck movement, when the eye position changed. The angle defined as thoracic bending became more acute, suggesting a more kyphotic thoracic spine, as vertical gaze direction and viewing angle were lowered. The inclination of the trunk and the pelvic angle correlated significantly with the viewing distance. The viewing distance seemed to be stabilized by the trunk to have a better view of the visual target. However, there appears to be no correlation between neck and trunk movement among the subjects in this experiment.

The relationship between neck movement and vertical gaze direction is rendered more distinct when considering the corresponding movement each made with respect to the horizontal in determining the viewing angle at each screen height setting. The ratio of changes in the neck angle with respect to the horizontal and changes in the eye inclination also with respect to the horizontal was taken at each screen height setting. Contribution of each parameter to produce the viewing angle was determined from this ratio. Mean ratio of neck angle and eye inclination significantly changed as the screen height was modified. The trend showed that the mean ratio increased as the screen height was raised (table 4). Less movement for one parameter meant more movement for the other. At higher screen height settings, the neck inclination was bigger than eye inclination. At lower screen height settings, eye inclination was more than neck inclination in producing the corresponding viewing angle (figure 6). Vertical gaze direction seemed to be limited at higher screen heights. Paired *t*-test

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Table 4. Mean ratio of neck angle and vertical gaze at different screen heights.

Screen height (n = 10)	|Neck angle| : |Vertical gaze|† (Mean ± SD)
80 cm	0·19 ± 0·15
90 cm	0·33 ± 0·21
100 cm	0·51 ± 0·29
110 cm	0·74 ± 0·26
120 cm	1·15 ± 0·66
For 50 observations	0·58 ± 0·49
Kruskal-Wallis	$p < 0.001$

†Absolute deviation of neck angle and vertical gaze direction from the horizontal.

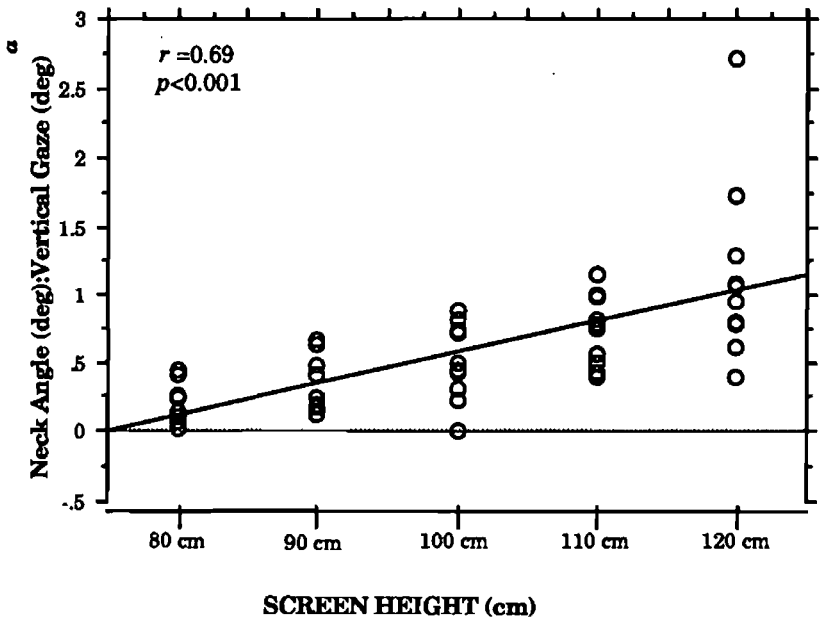


Figure 6. Relationship between neck angle to vertical gaze ratio and screen height.

showed that changes in the eye inclination from 100 to 110 cm and from 110 to 120 cm were not statistically significant (figure 7). This is in contrast with the changes in neck angle, which have been noted to be significant across all screen height settings (figure 8).

4. Discussion

The adaptation of individuals to changes in the workstation dimension will involve multiple movement of the anatomical elements. Lowering of the screen height of the VDT has brought about the lowering of the vertical gaze direction, decrease in ocular surface area and increase in neck flexion. The appropriate degree of downward eye position may seem favourable from the viewpoint of eye comfort. Yaginuma *et al.* (1990) and Tsubota and Nakamori (1993) reported that elevated viewing angle in VDT work resulted in reduced blinking rate and that the widening of the ocular

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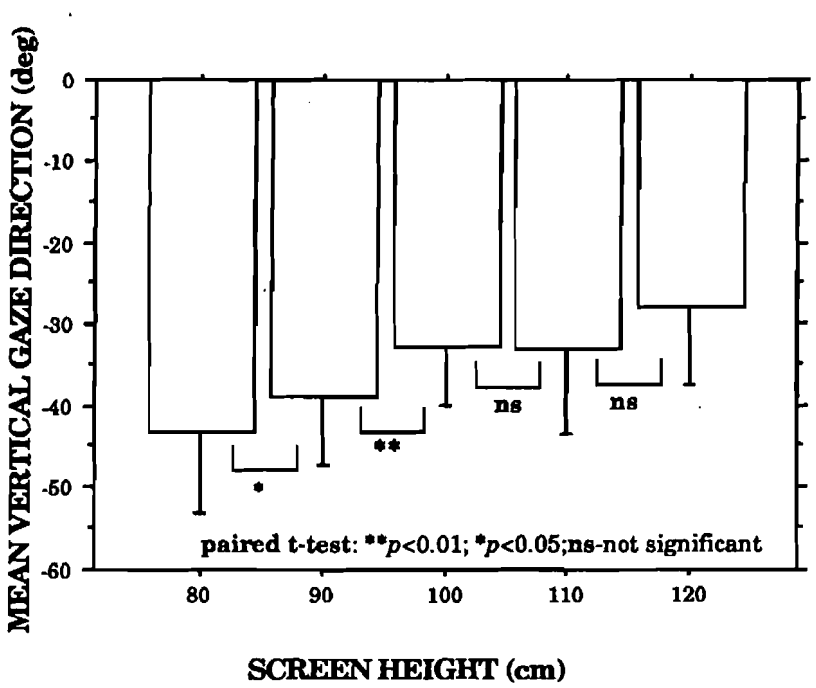


Figure 7. Mean vertical gaze direction at different screen heights.

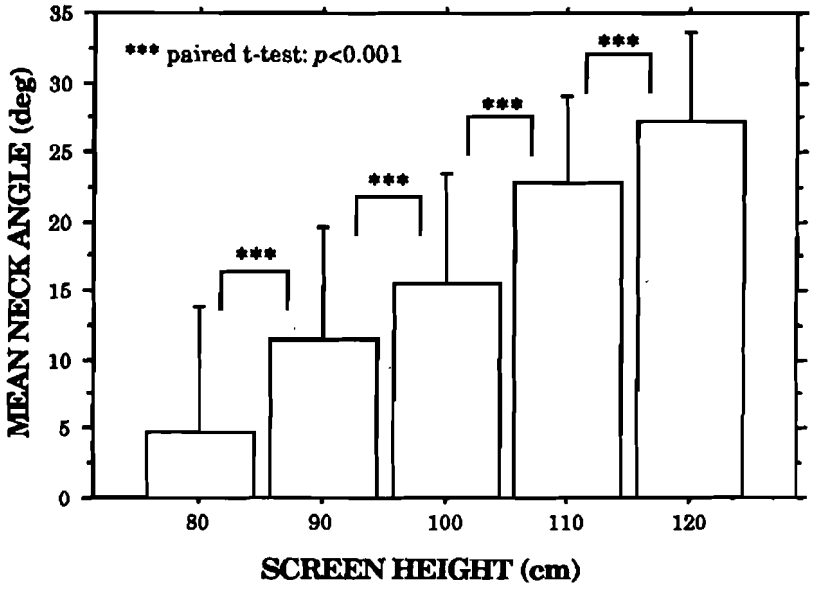


Figure 8. Mean neck angle at different screen heights.

surface area would consequently produce drying of the eyes and increased eye discomfort. Epidemiological findings of Bergqvist and Knave (1994) also showed that an elevated viewing angle was associated with eye discomfort. However, depression of gaze direction is invariably followed by increase in neck flexion. Many authors will contend that the flexed position of the neck may bring about

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unwanted consequences. Stress on the cervical spine is found to be higher in a flexed head as compared to a head assuming a more vertical position (Colombini *et al.* 1986, Schüldt *et al.* 1986). Fatigue also ensues earlier as the neck increases flexion (Chaffin 1973).

The relationship between the screen height and the trunk posture is also evident. The thoracic bending increased as the screen height was lowered. Throughout the experiment, the subjects were observed to assume a backward-leaning of the trunk. The meaning of these results, however, is not clear in view of an obscured relationship between back posture and discomfort. An erect sitting posture was associated with decreased discomfort (Sauter *et al.* 1991). On the other hand, EMG studies of the erector spinae muscle of the back showed that muscle activity was greater in an erect spine, suggesting more stress, than in a bent spine (Colombini *et al.* 1986). Moreover, a reclining posture is considered to reduce strain on the vertebral disc and back muscles (Andersson and Ortengreen 1974).

Direct comparison of parameters measured in this study to those measured by other authors cannot be made considering the differences in methodology. None the less, the relationship of the variables in this experiment is clear. The position of the eye is inevitably associated with a change in the neck angle and thoracic bending. Proponents of guidelines for downward vertical gaze direction will have to consider the potential burden on the neck and on the spine.

The trunk position maintained a backward-leaning position as the screen height was varied. The pelvic angle, likewise, was noted to remain almost unchanged. The seemingly immobile trunk and pelvis were also observed during VDT work in both experimental and actual work setting (Nakaseko *et al.* 1993, Miyamoto *et al.* 1995). Several factors may account for this constrained posture. The mouse table height and chair height were fixed for the duration of the experiment thereby imposing a constant work environment for the lower body. Another factor is the nature of the task. The game required constant visual monitoring by the subjects to be able to respond appropriately to the signal given by the computer. Further, the mouse was the only tool used as an input device of the computer. Handling of the mouse relied solely on tactile and spatial perception that eliminated the need to look down towards the mouse table. These factors present in this experimental situation may have produced visual fixation towards the screen. Body movements were simplified and concentrated only on the right hand, precluding the need to alter trunk and pelvic positions. Hence, after the preferred viewing angle has been decided and the corresponding neck position has been assumed, the posture of the trunk and pelvis were held almost constant.

The correlation matrix gave an important perspective concerning the relationship of body parts as they move when the workstation dimension is changed. From the results, posture is apparently decided by the demand on the visual system. Although no correlation between trunk inclination and screen height exists, an understanding of how the trunk functions to stabilize the viewing distance is evident. The neck appears to be the most susceptible and most important in adjusting its position to find the preferred viewing angle. The outcome of the experiment suggests that the neck inclination supports eye inclination especially at higher screen heights. Studies on eye physiology have demonstrated that the downward gaze leads to visual comfort during VDT work. The limited upward gaze at screen height greater than 100 cm, as seen in this study, may be taken as a way of preventing visual discomfort.

Ultimately, the accomplishment of work involving visual display terminals rests

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heavily on the human's visual system. Other movements of the body seem secondary and function to support the activities of the eye during work (Laville 1980, Colombini *et al.* 1986). The information pointing to priority given to eye position by the other body parts may provide direction in the amendment of existing guidelines.

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